EVS Project

♻ SMART WASTE CLASSIFICATION SYSTEM

AI-powered intelligent waste detection

Group Members:

Sanika Mane:202401110047

Sanskar Meharkar:202401110029

Mit Jangid:202401110072

INTRODUCTION

- Waste management is a major global issue
- Manual segregation is inefficient and time-consuming
- Need for automated smart systems
- AI can help classify waste accuratel

PROBLEM STATEMENT

- Improper waste segregation leads to pollution
- Recycling becomes difficult
- Human error in classification
- Lack of awareness and automation

OBJECTIVE

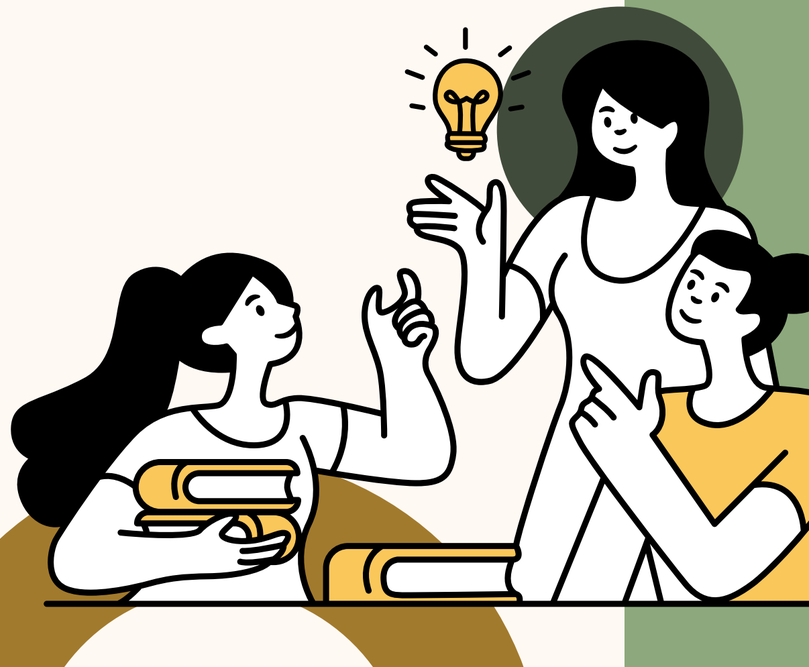
- To build an AI-based waste classification system
- Classify waste into different categories
- Improve recycling efficiency
- Provide user-friendly interface

DATASET

- Dataset collected from Kaggle
- Link:https://www.kaggle.com/datasets/techsash/waste-classification-data
- Categories include:

1. Plastic
2. Glass
3. Metal
4. Paper
5. Cardboard
6. Trash

- Images used for training and testing



TECHNOLOGIES USED

- Python
- TensorFlow / Keras
- Streamlit
- NumPy
- PIL (Image processing)

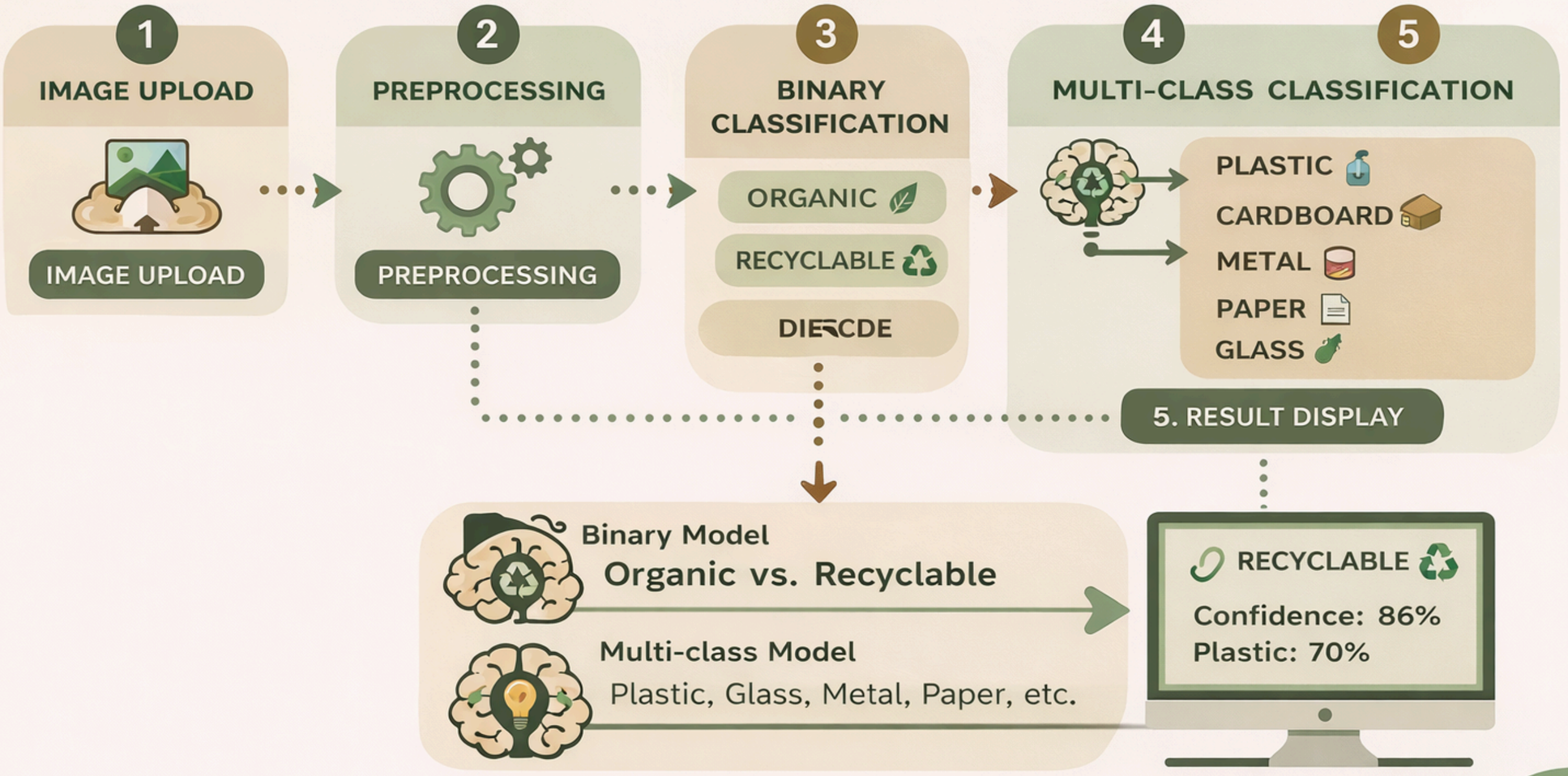
METHODOLOGY

- Image input from user
- Preprocessing (resize, normalize)
- Model prediction
- Display result with confidence

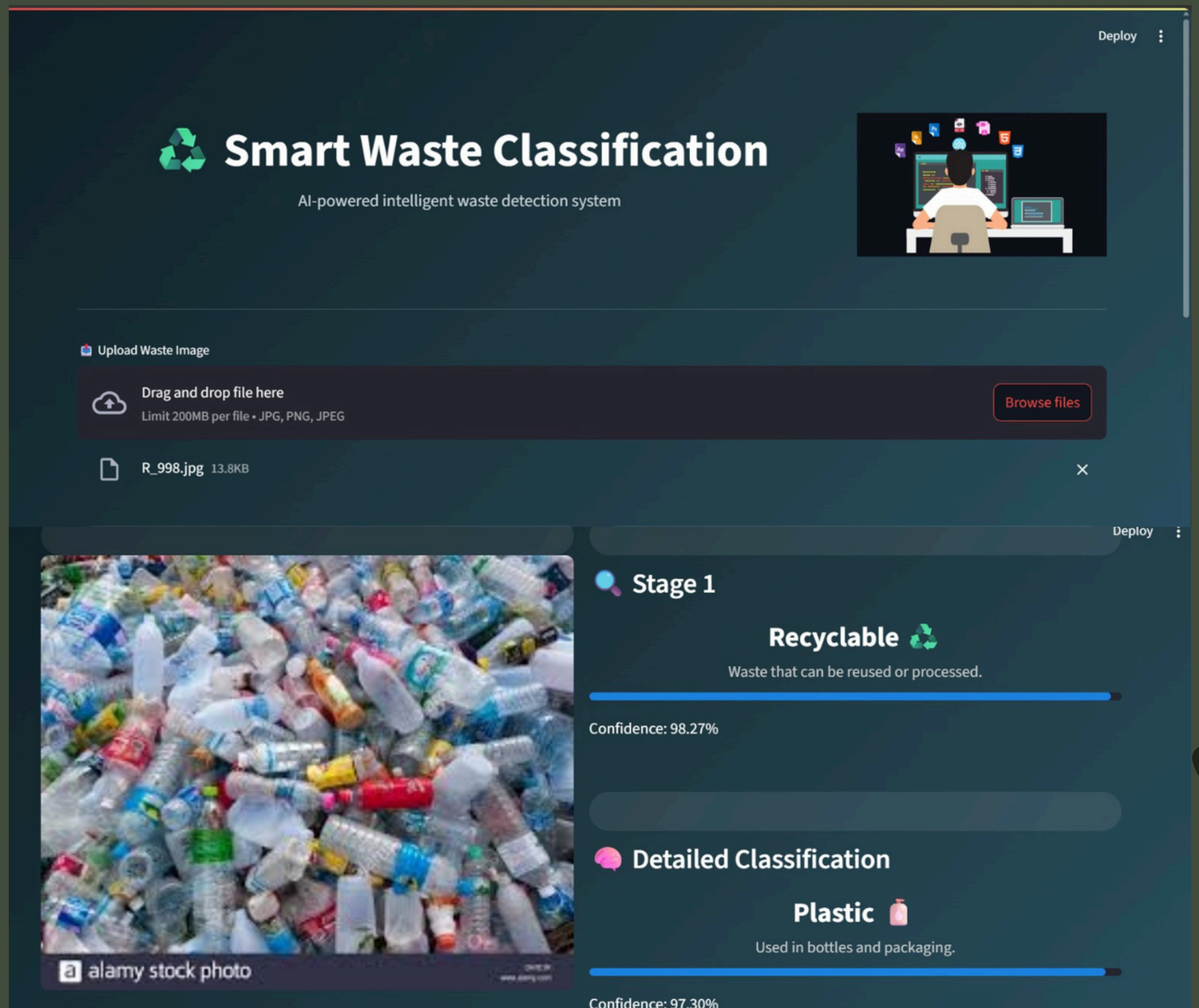
MODELS USED

- Binary Classification Model:
  - Organic vs Recyclable
- Multi-class Classification Model:
  - Plastic, Glass, Metal, Paper, etc.
- CNN (Convolutional Neural Network) used

SYSTEM ARCHITECTURE



OUTPUT / RESULTS



- Displays predicted class
- Shows confidence percentage
- Example:
  - Recyclable → Plastic
- Fast and user-friendly

ADVANTAGES

- Reduces human effort
- Improves recycling process
- Fast and accurate
- Easy to use interface

FUTURE SCOPE

- Deploy in smart cities
- Real-time camera integration
- Mobile application
- Improve model accuracy

CONCLUSION

- AI can significantly improve waste management
- System helps in proper waste segregation
- Contributes to environmental sustainability

THANK YOU

Project Github Link:

https://github.com/Sanika13112/Smart-Waste-Classification: